

Azure Ubuntu Deployment Steps

Step-by-step deployment guide for the React frontend, FastAPI backend, and PostgreSQL database. Keep your private key private; do not paste it into chats or documentation.

1. SSH Into The VM

From your local machine, connect to the Azure Ubuntu VM using your private key and the VM public IP address. Replace the username if your VM uses a different one.

```
ssh -i /path/to/private-key.pem azureuser@YOUR_VM_PUBLIC_IP
```

2. Update Ubuntu

```
sudo apt update
sudo apt upgrade -y
```

3. Install Required Packages

```
sudo apt install -y python3 python3-venv python3-pip postgresql postgresql-contrib nginx git
```

4. Create PostgreSQL Database And User

Create a PostgreSQL database for the inventory app. Use a strong password and store it safely.

```
sudo -u postgres psql
CREATE DATABASE invent;
CREATE USER invent_user WITH PASSWORD 'YOUR_SECURE_PASSWORD';
GRANT ALL PRIVILEGES ON DATABASE invent TO invent_user;
\q
```

5. Get The Project Onto The VM

Use one of these approaches. Git is cleaner if your project is in a repository. SCP is useful if the project is only on your local machine.

```
git clone YOUR_REPO_URL
# or upload from local machine:
scp -i /path/to/private-key.pem -r ./invent azureuser@YOUR_VM_PUBLIC_IP:/home/azureuser/invent
```

6. Configure The FastAPI Backend

```
cd ~/invent/backend
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
cp .env.example .env
```

Edit backend/.env and set your real PostgreSQL connection string and frontend origin.

```
DATABASE_URL=postgresql+psycopg://invent_user:YOUR_SECURE_PASSWORD@localhost:5432/invent
BACKEND_CORS_ORIGINS=http://YOUR_VM_PUBLIC_IP,http://YOUR_DOMAIN
```

7. Test FastAPI Manually

```
cd ~/invent/backend
source .venv/bin/activate
uvicorn app.main:app --host 0.0.0.0 --port 8000
```

- Temporarily allow Azure inbound port 8000 only for testing.
- Visit `http://YOUR_VM_PUBLIC_IP:8000/health` and confirm it returns `{"status":"ok"}`.
- After Nginx reverse proxy is configured, close public port 8000.

8. Create A Systemd Service For FastAPI

Create a service so the backend starts automatically after reboot.

```
sudo nano /etc/systemd/system/invent-api.service
```

```
[Unit]
Description=Invent FastAPI Backend
After=network.target

[Service]
User=azureuser
WorkingDirectory=/home/azureuser/invent/backend
Environment="PATH=/home/azureuser/invent/backend/.venv/bin"
ExecStart=/home/azureuser/invent/backend/.venv/bin/uvicorn app.main:app --host 127.0.0.1 --port 8000
Restart=always

[Install]
WantedBy=multi-user.target
```

```
sudo systemctl daemon-reload
sudo systemctl enable invent-api
sudo systemctl start invent-api
sudo systemctl status invent-api
```

9. Build The React Frontend

Set the frontend API URL before building. If you use Nginx reverse proxy, use `/api` or your public backend path.

```
cd ~/invent
npm install
nano .env
VITE_API_BASE_URL=http://YOUR_VM_PUBLIC_IP/api
npm run build
```

10. Configure Nginx

Nginx should serve the React dist folder and reverse proxy API calls to FastAPI running locally on `127.0.0.1:8000`.

```
sudo nano /etc/nginx/sites-available/invent
```

```
server {
    listen 80;
    server_name YOUR_VM_PUBLIC_IP YOUR_DOMAIN;

    root /home/azureuser/invent/dist;
    index index.html;

    location / {
        try_files $uri /index.html;
    }

    location /api/ {
        proxy_pass http://127.0.0.1:8000/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

```
sudo ln -s /etc/nginx/sites-available/invent /etc/nginx/sites-enabled/invent
sudo nginx -t
sudo systemctl reload nginx
```

11. Configure Azure Network Security Group

- Allow port 22 for SSH. Restrict this to your IP if possible.
- Allow port 80 for HTTP.
- Allow port 443 after HTTPS is configured.
- Close port 8000 after Nginx proxies /api to FastAPI.
- Do not expose PostgreSQL publicly. It should stay on localhost only.

12. Add HTTPS Later

After your domain points to the VM public IP, install Certbot and issue a Let's Encrypt certificate.

```
sudo apt install -y certbot python3-certbot-nginx
sudo certbot --nginx -d YOUR_DOMAIN
```

Final Target Architecture

```
Browser -> Nginx :80/:443 -> React dist
Browser -> Nginx /api -> FastAPI on 127.0.0.1:8000
FastAPI -> PostgreSQL on localhost:5432
```

Important Notes

- Never commit .env files or private keys.
- Use PostgreSQL as the primary database.
- Keep frontend localStorage only as a temporary cache/fallback.
- Add database backups before real production use.
- Use HTTPS before entering sensitive inventory or user data.